

Wildcard Paper

Disclaimer: This is a pattern-based predicted practice paper workflow output, not a leaked, exact, or official CAT paper.

Variant Blueprint

- Variant type: wildcard
- Strategy: Favor diversity and volatility.
- Target section mix: QA 22 from the frozen selected pool; VARC/DILR are outside the current candidate scope
- Expected overlap score: 0.817
- Risk score: 0.65
- Diversity score: 0.636

Questions

Q1. QA - Geometry / Triangles

- Archetype/type: Triangle similarity problem
- Difficulty: Medium-Hard
- Score: 0.865

In triangle ABC , point D lies on side AB and point E lies on side AC such that $DE \parallel BC$. The line through D parallel to AC meets BC at point F . If the area of triangle ADE is 16 and the area of quadrilateral $DECF$ is 40, find the area of triangle DBF .

A. 20 B. 24 C. 25 D. 36

Q2. QA - Geometry / Coordinate Geometry

- Archetype/type: Coordinate geometry line/slope/distance problem
- Difficulty: Hard
- Score: 0.839

In a parallelogram $ABCD$ with vertices taken in that order, $A = (1, 2)$ and $B = (7, 5)$. The vertex $C = (p, q)$ has integer coordinates and $p > 10$. The diagonals intersect at a point lying on the line $x + 2y = 16$, and the area of the parallelogram is 42. Find $p + q$.

Q3. QA - Geometry / Mensuration

- Archetype/type: Mensuration volume/area problem
- Difficulty: Hard
- Score: 0.811

A right circular cone has base radius 10 cm and height 24 cm. It is cut by a plane parallel to the base so that the curved surface area of the smaller cone

removed from the top is one-fourth of the curved surface area of the original cone. The remaining frustum is melted and recast into identical solid cylinders, each of radius 5 cm and height 7 cm. How many such cylinders can be made?

A. 3 B. 4 C. 5 D. 6

Q4. QA - Number System / Divisibility

- Archetype/type: Divisibility/factor-counting problem
- Difficulty: Hard
- Score: 0.808

Find the least positive integer n such that n leaves remainders 3, 5, and 7 when divided by 5, 7, and 9, respectively, and n is divisible by 11.

A. 943 B. 1258 C. 1573 D. 1888

Q5. QA - Number System / Digits

- Archetype/type: Digit manipulation problem
- Difficulty: Hard
- Score: 0.727

How many three-digit numbers have distinct non-zero digits and become larger by 198 when their digits are reversed?

A. 35 B. 42 C. 49 D. 56

Q6. QA - Arithmetic / Profit and Loss

- Archetype/type: Profit-loss transaction problem
- Difficulty: Hard
- Score: 0.91

A shop buys 90 identical bags at Rs 640 each. It marks every bag at the same price. Of the 90 bags, 36 are sold at a 10% discount, 30 at a 20% discount, 12 at a 35% discount because of minor defects, and 12 remain unsold. If the shop still makes an overall profit of Rs 6600 on the entire purchase, what is the marked price of each bag?

A. Rs 900 B. Rs 960 C. Rs 1000 D. Rs 1080

Q7. QA - Arithmetic / Mixtures and Alligation

- Archetype/type: Mixture concentration problem
- Difficulty: Hard
- Score: 0.905

Solution A is 40% acid by volume and Solution B is 70% acid by volume. A chemist mixes a litres of Solution A with b litres of Solution B. After mixing,

exactly 10 litres of pure water evaporates from the combined mixture. The remaining mixture has volume 40 litres and is 75% acid. Find the ratio $a : b$.

A. 1 : 1 B. 1 : 2 C. 2 : 3 D. 3 : 5

Q8. QA - Arithmetic / Percentages

- Archetype/type: Successive percentage change
- Difficulty: Hard
- Score: 0.892

A value is first increased by $p\%$, then decreased by 20%, and then again increased by $p\%$. The final value is exactly 90% of what the value would have been after only a single $p\%$ increase from the original. If p is positive, find p .

Q9. QA - Arithmetic / Ratio and Proportion

- Archetype/type: Ratio sharing problem
- Difficulty: Hard
- Score: 0.873

The monthly incomes of P and Q are in the ratio 8:11, while their monthly expenditures are in the ratio 5:7. Q saves Rs 3000 more than P. After P invests 20% of his saving and Q invests 25% of his saving, their remaining savings are equal. What is Q's monthly income in rupees?

Q10. QA - Arithmetic / Time and Work

- Archetype/type: Work-efficiency problem
- Difficulty: Hard
- Score: 0.862

A and B together can finish a job in 16 days, B and C together in 24 days, and A and C together in 18 days. A starts alone. After 4 days, C joins. After 3 more days, B also joins. How many days from the start are needed to finish the job?

A. 15.25 days B. 357/23 days C. 16 days D. 371/24 days

Q11. QA - Arithmetic / Averages

- Archetype/type: Weighted average problem
- Difficulty: Hard
- Score: 0.859

In a batch of 40 candidates, the average score is 69.6. The highest 5 candidates have an average score of 92, and the lowest 7 candidates have an average score of 48. The lowest 7 candidates are replaced by 7 new candidates. If the new average score of all 40 candidates becomes 74.5, find the average score of the 7 new candidates.

Q12. QA - Algebra / Inequalities

- Archetype/type: Inequality optimization problem
- Difficulty: Hard
- Score: 0.855

For integer k with $0 \leq k \leq 15$, how many values of k make both roots of

$$x^2 + (k - 3)x + (k - 2) = 0$$

distinct negative real numbers?

A. 7 B. 8 C. 9 D. 12

Q13. QA - Algebra / Linear Equations

- Archetype/type: Integer feasibility under hidden divisibility constraint
- Difficulty: Hard
- Score: 0.854

A company has three departments X, Y, and Z. Every employee belongs to exactly one department. The number of employees in X is a positive multiple of 4. The number of employees in Y exceeds that in X by a positive multiple of 3. The number of employees in Z equals exactly half the sum of the employees in X and Y. If the total number of employees in the company is between 100 and 130, inclusive, how many distinct values can the total number of employees take?

Q14. QA - Algebra / Functions

- Archetype/type: Function value/domain problem
- Difficulty: Hard
- Score: 0.852

A function f defined on positive integers satisfies $f(1) = 1$ and

$$f(n) = n - f(f(n - 1)) \quad \text{for } n \geq 2.$$

Find $f(20)$.

A. 10 B. 11 C. 12 D. 13

Q15. QA - Geometry / Triangles

- Archetype/type: Triangle similarity problem
- Difficulty: Hard
- Score: 0.835

In triangle ABC , points P and Q lie on sides AB and AC respectively such that $PQ \parallel BC$. The line through P parallel to AC meets BC at R , and the

line through Q parallel to AB meets BC at S . It is given that $[APQ] = 25$, $[PBR] = 9$, and $PQ < BC$. Find the area of triangle ABC .

A. 34 B. 49 C. 64 D. 100

Q16. QA - Number System / Divisibility

- Archetype/type: Divisibility/factor-counting problem
- Difficulty: Hard
- Score: 0.796

How many five-digit numbers of the form $3a7b2$, where a and b are digits, are divisible by 18 but not by 36?

A. 4 B. 5 C. 6 D. 7

Q17. QA - Geometry / Mensuration

- Archetype/type: Mensuration volume/area problem
- Difficulty: Hard
- Score: 0.793

A solid capsule is formed by attaching two hemispheres to the two ends of a right circular cylinder. The cylinder and the hemispheres have the same radius r , and the cylindrical part has height h . If every linear dimension of the capsule is doubled, its surface area increases by 312π and its volume increases by $\frac{3472\pi}{3}$. Find $h - r$.

A. 1 B. 2 C. 3 D. 4

Q18. QA - Geometry / Coordinate Geometry

- Archetype/type: Coordinate geometry line/slope/distance problem
- Difficulty: Hard
- Score: 0.789

In parallelogram $ABCD$, the vertices are in that order. Let $A = (0, 1)$ and $B = (6, 4)$. The point $C = (p, q)$ has integer coordinates with $q < 5$. The diagonals intersect at a point lying on the line $2x + y = 21$, and the area of the parallelogram is 45. Find $p + q$.

Q19. QA - Geometry / Triangles

- Archetype/type: Triangle similarity problem
- Difficulty: Hard
- Score: 0.76

In triangle ABC , D is the midpoint of AB . A point E lies on AC such that DE is extended to meet the line through C parallel to AB at F . If $AE:EC = 2:1$ and area of triangle ADE is 18, what is the area of triangle CEF ?

A. 4.5 B. 6 C. 9 D. 12

Q20. QA - Geometry / Mensuration

- Archetype/type: Mensuration volume/area problem
- Difficulty: Hard
- Score: 0.698

A rectangle is enlarged by increasing its length by 25% and decreasing its breadth by 10%. The enlarged rectangle is then cut into three equal-area parts. Each part has area 45 square cm. What was the original area?

A. 108 sq cm B. 120 sq cm C. 135 sq cm D. 150 sq cm

Q21. QA - Number System / Divisibility

- Archetype/type: Divisibility/factor-counting problem
- Difficulty: Hard
- Score: 0.678

How many integers from 1 to 600 are divisible by exactly one of 6, 10 and 15, but not by all three? The phrase exactly one is to be interpreted literally.

A. 120 B. 130 C. 140 D. 150

Q22. QA - Number System / Divisibility

- Archetype/type: Divisibility/factor-counting problem
- Difficulty: Hard
- Score: 0.711

How many integers from 1 to 300 are divisible by exactly one of 4, 6 and 10?

A. 60 B. 70 C. 80 D. 90

Answer Key

Q	Answer	Score
1	(C) 25	0.865
2	23	0.839
3	(B) 4	0.811
4	(C) 1573	0.808
5	49	0.727
6	Rs 1000	0.91
7	(B) 1 : 2	0.905
8	12.5	0.892
9	825000	0.873
10	357/23 days	0.862

Q	Answer	Score
11	76	0.859
12	(B) 8	0.855
13	10	0.854
14	(C) 12	0.852
15	(C) 64	0.835
16	(C) 6	0.796
17	(A) 1	0.793
18	22	0.789
19	4.5	0.76
20	120 sq cm	0.698
21	140	0.678
22	70	0.711

Detailed Solutions

Q1

Step 1: Let $\frac{AD}{AB} = k$. Since $DE \parallel BC$, triangle $ADE \sim ABC$ with linear ratio k , so $[ADE] = k^2 \cdot [ABC]$. Step 2: Since $DF \parallel AC$, triangle $DBF \sim ABC$ (same angle at B , with $DB/AB = 1 - k$ and $BF/BC = 1 - k$). Hence $[DBF] = (1 - k)^2 \cdot [ABC]$. Step 3: Identify the hidden structure of $DECF$. Both $DE \parallel BC$ and $DF \parallel AC$. By the proportionality theorem, $DE = k \cdot BC$ and $FC = BC - BF = (1 - (1 - k)) \cdot BC = k \cdot BC$. So $DE = FC$ with $DE \parallel FC$, meaning $DECF$ is a *parallelogram*, not just a trapezoid. Step 4: Since the three regions $[ADE]$, $[DECF]$, $[DBF]$ partition $[ABC]$:

$$[ABC] = [ADE] + [DBF] + [DECF] = k^2 T + (1 - k)^2 T + 2k(1 - k)T = T.$$

So $[DECF] = 2k(1 - k)T$. Step 5: Now $[DECF]^2 = 4k^2(1 - k)^2 T^2 = 4 \cdot [ADE] \cdot [DBF]$. Substituting:

$$40^2 = 4 \cdot 16 \cdot [DBF] \Rightarrow [DBF] = \frac{1600}{64} = 25.$$

Q2

Step 1: In a parallelogram, the diagonals bisect each other. Hence the midpoint of AC is

$$\left(\frac{1 + p}{2}, \frac{2 + q}{2} \right).$$

Step 2: This point lies on $x + 2y = 16$, so

$$\frac{1 + p}{2} + 2 \left(\frac{2 + q}{2} \right) = 16.$$

Step 3: Simplify:

$$\frac{1+p}{2} + 2 + q = 16.$$

Multiplying by 2,

$$1 + p + 4 + 2q = 32,$$

so

$$p + 2q = 27.$$

Step 4: The adjacent side vectors are

$$\overrightarrow{AB} = (6, 3), \quad \overrightarrow{BC} = (p - 7, q - 5).$$

The area of the parallelogram is the absolute value of their determinant:

$$|6(q - 5) - 3(p - 7)| = 42.$$

Step 5: Substitute $p = 27 - 2q$:

$$|6q - 30 - 3(27 - 2q - 7)| = 42.$$

This becomes

$$|6q - 30 - 3(20 - 2q)| = 42,$$

or

$$|12q - 90| = 42.$$

Step 6: Therefore,

$$12q - 90 = 42 \quad \text{or} \quad 12q - 90 = -42.$$

So

$$q = 11 \quad \text{or} \quad q = 4.$$

Step 7: From $p + 2q = 27$, these give

$$(q, p) = (11, 5) \quad \text{or} \quad (4, 19).$$

The condition $p > 10$ selects

$$p = 19, \quad q = 4.$$

Hence

$$p + q = 23.$$

Q3

Step 1: The original cone has radius 10 and height 24, so its slant height is

$$\sqrt{10^2 + 24^2} = 26.$$

Step 2: The curved surface area of a cone is proportional to rl . For similar cones, both r and l scale by the same linear factor, so curved surface area scales as the square of the linear factor. Step 3: Since the smaller cone has one-fourth the curved surface area of the original cone, its linear scale factor is

$$\sqrt{\frac{1}{4}} = \frac{1}{2}.$$

Step 4: Therefore the volume of the smaller cone is

$$\left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

of the original cone's volume. Step 5: The original cone's volume is

$$\frac{1}{3}\pi(10)^2(24) = 800\pi.$$

So the frustum volume is

$$800\pi - \frac{1}{8}(800\pi) = 700\pi.$$

Step 6: Each cylinder has volume

$$\pi(5)^2(7) = 175\pi.$$

Therefore the number of cylinders is

$$\frac{700\pi}{175\pi} = 4.$$

Q4

Step 1: The given remainder conditions are

$$n \equiv 3 \pmod{5}, \quad n \equiv 5 \pmod{7}, \quad n \equiv 7 \pmod{9}.$$

Step 2: Each remainder is 2 less than the corresponding divisor. Hence

$$n + 2$$

is divisible by 5, 7, and 9. Step 3: Since

$$\text{lcm}(5, 7, 9) = 315,$$

we can write

$$n + 2 = 315k,$$

so

$$n = 315k - 2.$$

Step 4: Now impose divisibility by 11:

$$315k - 2 \equiv 0 \pmod{11}.$$

Step 5: Since

$$315 \equiv 7 \pmod{11},$$

we get

$$7k - 2 \equiv 0 \pmod{11},$$

or

$$7k \equiv 2 \pmod{11}.$$

Step 6: The inverse of 7 modulo 11 is 8, because

$$7 \cdot 8 = 56 \equiv 1 \pmod{11}.$$

Therefore,

$$k \equiv 2 \cdot 8 = 16 \equiv 5 \pmod{11}.$$

Step 7: The least positive value is $k = 5$. Hence

$$n = 315(5) - 2 = 1575 - 2 = 1573.$$

Q5

Step 1: Let the number be $100a+10b+c$. Step 2: Reversed-original equals $99(c-a)$. Step 3: Given 198, $c-a=2$. Step 4: a can be 1 through 7. Step 5: For each, b has 7 non-zero choices excluding a and c , total 49.

Q6

Step 1: Total cost of all 90 bags = $90 \times 640 = \text{Rs } 57600$. Step 2: Profit is calculated on the entire purchase, including the unsold bags, so required revenue = $57600 + 6600 = \text{Rs } 64200$. Step 3: Let the marked price be M . Only 78 bags are sold, giving revenue $36 \times 0.90M + 30 \times 0.80M + 12 \times 0.65M = 32.4M + 24M + 7.8M = 64.2M$. Step 4: Therefore $64.2M = 64200$, so $M = \text{Rs } 1000$. Step 5: Verification: revenue = $36 \times 900 + 30 \times 800 + 12 \times 650 = 32400 + 24000 + 7800 = \text{Rs } 64200$.

Q7

Step 1: After evaporation, the remaining volume is 40 litres and the acid concentration is 75%. Hence acid in the final mixture is

$$0.75 \times 40 = 30 \text{ litres.}$$

Step 2: Evaporation removes only water, not acid. Therefore acid before evaporation was also 30 litres. Step 3: Since 10 litres of water evaporated and final volume is 40 litres, the pre-evaporation mixed volume was

$$40 + 10 = 50 \text{ litres.}$$

So

$$a + b = 50.$$

Step 4: Acid balance before evaporation:

$$0.40a + 0.70b = 30.$$

Step 5: Substitute $a = 50 - b$:

$$0.40(50 - b) + 0.70b = 30,$$

$$20 - 0.40b + 0.70b = 30,$$

$$0.30b = 10 \Rightarrow b = \frac{100}{3}.$$

Then

$$a = 50 - \frac{100}{3} = \frac{50}{3}.$$

Step 6: Therefore

$$a : b = \frac{50}{3} : \frac{100}{3} = 1 : 2.$$

Q8

Step 1: Let the original value be 100. Step 2: After the three-stage change, the value is $100(1+p/100)(0.8)(1+p/100)$. Step 3: After only a single $p\%$ increase, the value would be $100(1+p/100)$. Step 4: The first value is 90% of the second, so $0.8(1+p/100)^2 = 0.9(1+p/100)$. Step 5: Since p is positive, divide by the common positive factor to get $0.8(1+p/100)=0.9$, so $p=12.5$.

Q9

Step 1: Let incomes be $8x$ and $11x$, and expenditures be $5y$ and $7y$. Then P's saving is $8x - 5y$ and Q's saving is $11x - 7y$. Step 2: Q saves Rs 3000 more, so $11x - 7y = 8x - 5y + 3000$, giving $3x - 2y = 3000$. Step 3: After investment, remaining savings are 80% of P's saving and 75% of Q's saving, so $0.8(8x - 5y) = 0.75(11x - 7y)$. Since Q's saving exceeds P's by 3000, this is also $0.8(SQ - 3000) = 0.75SQ$. Hence $SQ = 48000$. Step 4: Therefore $11x - 7y = 48000$ and $3x - 2y = 3000$. From the second equation, $y = 1.5x - 1500$. Step 5: Substitute: $11x - 7(1.5x - 1500) = 48000$, so $0.5x + 10500 = 48000$, $x = 75000$. Q's income = $11x = \text{Rs } 825000$.

Q10

Let the daily rates of A, B and C be a , b and c . We have $a+b = 1/16$, $b+c = 1/24$, and $a+c = 1/18$. Adding gives $2(a+b+c) = 1/16 + 1/24 + 1/18 = 9/144 + 6/144 + 8/144 = 23/144$, so $a+b+c = 23/288$. Also $a = [(a+b)+(a+c)-(b+c)]/2 = (1/16 + 1/18 - 1/24)/2 = (9+8-6)/144$ divided by 2 = $11/288$. Similarly $c = [(b+c)+(a+c)-(a+b)]/2 = (6+8-9)/144$ divided by 2 = $5/288$. In the first 4 days A completes $4 \times 11/288 = 44/288 = 11/72$. In the next 3 days A+C complete $3 \times (16/288) = 48/288 = 1/6$. Work done in first 7 days = $11/72 + 1/6 = 23/72$. Remaining work = $49/72$. With all three working, time needed = $(49/72)/(23/288) = 196/23$ days. Total time = $7 + 196/23 = 357/23$ days.

Q11

Step 1: Convert the original average into the original total:

$$40 \times 69.6 = 2784.$$

Step 2: The total score of the lowest 7 candidates is

$$7 \times 48 = 336.$$

Step 3: After replacing these 7 candidates, the total score of the unchanged 33 candidates is

$$2784 - 336 = 2448.$$

Step 4: The new average of all 40 candidates is 74.5, so the new total score is

$$40 \times 74.5 = 2980.$$

Step 5: Therefore, the total score of the 7 new candidates is

$$2980 - 2448 = 532.$$

Step 6: Hence their average score is

$$\frac{532}{7} = 76.$$

Q12

Step 1: For both roots negative and *distinct* real:

- Discriminant > 0 (strict, for distinct): $(k-3)^2 - 4(k-2) > 0$.
- Sum of roots negative: $-(k-3) < 0$, i.e. $k > 3$.
- Product of roots positive: $k-2 > 0$, i.e. $k > 2$.

Step 2: Simplify discriminant: $(k-3)^2 - 4(k-2) = k^2 - 6k + 9 - 4k + 8 = k^2 - 10k + 17 > 0$. Step 3: Roots of $k^2 - 10k + 17 = 0$: $k = \frac{10 \pm \sqrt{100 - 68}}{2} =$

$$\frac{10 \pm \sqrt{32}}{2} = 5 \pm 2\sqrt{2}. \text{ Approximately } 5 + 2\sqrt{2} \approx 7.83 \text{ and } 5 - 2\sqrt{2} \approx 2.17.$$

Step 4: So $k^2 - 10k + 17 > 0$ iff $k < 5 - 2\sqrt{2}$ or $k > 5 + 2\sqrt{2}$. Step 5: Combine with $k > 3$ and $k > 2$: intersection is $k > 5 + 2\sqrt{2} \approx 7.83$ (the branch $k < 5 - 2\sqrt{2} \approx 2.17$ is killed by $k > 3$). Step 6: Integer k with $7.83 < k \leq 15$: $k \in \{8, 9, 10, 11, 12, 13, 14, 15\}$. Count = 8. Step 7: Verify endpoints. $k = 8$: $x^2 + 5x + 6 = (x + 2)(x + 3) = 0$, roots $-2, -3$, both negative and distinct $\checkmark$. $k = 7$: $x^2 + 4x + 5 = 0$, discriminant $= 16 - 20 = -4 < 0$, complex roots $\times$ (good).

Q13

Step 1: Let the number of employees in X be $4a$ where $a \geq 1$, and let Y exceed X by $3b$ where $b \geq 1$. Then $|Y| = 4a + 3b$. Step 2: The number of employees in Z is

$$|Z| = \frac{|X| + |Y|}{2} = \frac{4a + 4a + 3b}{2} = \frac{8a + 3b}{2}.$$

For $|Z|$ to be a positive integer, $8a + 3b$ must be even. Since $8a$ is always even, $3b$ must be even, which forces b to be even. Write $b = 2k$ with $k \geq 1$. Step 3: With $b = 2k$, we get $|Z| = 4a + 3k$ and

$$T = |X| + |Y| + |Z| = 4a + (4a + 6k) + (4a + 3k) = 12a + 9k = 3(4a + 3k).$$

So T is always a multiple of 3. Step 4: Impose $100 \leq T \leq 130$:

$$100 \leq 3(4a + 3k) \leq 130 \implies 34 \leq 4a + 3k \leq 43.$$

Let $m = 4a + 3k$ where $a \geq 1$ and $k \geq 1$, so $m \geq 7$. For any integer $m \geq 7$, a representation $m = 4a + 3k$ with $a, k \geq 1$ exists: if $m \equiv 1 \pmod{3}$, take $a = 1$; if $m \equiv 2 \pmod{3}$, take $a = 2$; if $m \equiv 0 \pmod{3}$, take $a = 3$; and solve for k accordingly. Each resulting k is positive for $m \geq 7$. Step 5: The integers m from 34 to 43 inclusive give $43 - 34 + 1 = 10$ values of m , hence 10 distinct totals $T \in \{102, 105, 108, 111, 114, 117, 120, 123, 126, 129\}$.

Q14

Step 1: Compute $f(n)$ step by step. $f(1) = 1$. $f(2) = 2 - f(f(1)) = 2 - f(1) = 1$. $f(3) = 3 - f(f(2)) = 3 - f(1) = 2$. $f(4) = 4 - f(f(3)) = 4 - f(2) = 3$. $f(5) = 5 - f(f(4)) = 5 - f(3) = 3$. $f(6) = 6 - f(f(5)) = 6 - f(3) = 4$. $f(7) = 7 - f(f(6)) = 7 - f(4) = 4$. $f(8) = 8 - f(f(7)) = 8 - f(4) = 5$. Step 2: Continue. $f(9) = 9 - f(f(8)) = 9 - f(5) = 6$. $f(10) = 10 - f(f(9)) = 10 - f(6) = 6$. $f(11) = 11 - f(f(10)) = 11 - f(6) = 7$. $f(12) = 12 - f(f(11)) = 12 - f(7) = 8$. Step 3: Continue. $f(13) = 13 - f(f(12)) = 13 - f(8) = 8$. $f(14) = 14 - f(f(13)) = 14 - f(8) = 9$. $f(15) = 15 - f(f(14)) = 15 - f(9) = 9$. $f(16) = 16 - f(f(15)) = 16 - f(9) = 10$. Step 4: Continue. $f(17) = 17 - f(f(16)) = 17 - f(10) = 11$. $f(18) = 18 - f(f(17)) = 18 - f(11) = 11$. $f(19) = 19 - f(f(18)) = 19 - f(11) = 12$. $f(20) = 20 - f(f(19)) = 20 - f(12) = 20 - 8 = 12$. Step 5: $f(20) = 12$.

Q15

Step 1: Let $k = \frac{AP}{AB}$. Since $PQ \parallel BC$, triangle $APQ \sim ABC$ with ratio k , so

$$[APQ] = k^2 \cdot [ABC] = 25.$$

Step 2: Since $PR \parallel AC$, triangle PBR has $\angle B$ common with ABC , with $PB/AB = 1 - k$ and $BR/BC = 1 - k$ (by basic proportionality on the parallel line PR to AC). Hence $PBR \sim ABC$ with ratio $1 - k$, giving

$$[PBR] = (1 - k)^2 \cdot [ABC] = 9.$$

Step 3: Divide the two equations: $\frac{k^2}{(1 - k)^2} = \frac{25}{9}$, so $\frac{k}{1 - k} = \pm \frac{5}{3}$. Since $k \in (0, 1)$, both $k/(1 - k) = 5/3$ (giving $k = 5/8$) and $k/(1 - k) = -5/3$ (giving $k = -5/2$, infeasible) need to be checked. Only $k = 5/8$ remains. Step 4: The condition $PQ < BC$ translates to $k < 1$, which is automatically true. (If the problem instead gave $[PBR]$ and $[QCS]$ separately and asked for two scenarios, the symmetric ambiguity in $k \leftrightarrow 1 - k$ would matter. Here $k = 5/8$ is unique.) Step 5: Substituting back: $[ABC] = \frac{25}{k^2} = \frac{25}{25/64} = 64$. Step 6: Verification. With $[ABC] = 64$, $k = 5/8$: $[APQ] = (25/64) \cdot 64 = 25 \checkmark$; $[PBR] = (9/64) \cdot 64 = 9 \checkmark$.

Q16

Step 1: Since the last digit is 2, the number is even. Thus divisibility by 18 reduces to divisibility by 9 along with evenness. Step 2: The digit sum is

$$3 + a + 7 + b + 2 = 12 + a + b.$$

Step 3: For divisibility by 9,

$$12 + a + b$$

must be a multiple of 9. Since $a + b$ ranges from 0 to 18, the possible values are

$$a + b = 6 \quad \text{or} \quad a + b = 15.$$

Step 4: Since $36 = 4 \times 9$, a number already divisible by 9 is divisible by 36 precisely when its last two digits form a multiple of 4. Step 5: The last two digits are $b2$, representing $10b + 2$. Modulo 4,

$$10b + 2 \equiv 2b + 2 \pmod{4}.$$

This is divisible by 4 when b is odd. Therefore, not divisible by 36 requires b to be even. Step 6: For $a + b = 6$ with b even, the valid pairs are

$$(b, a) = (0, 6), (2, 4), (4, 2), (6, 0),$$

giving 4 pairs. Step 7: For $a + b = 15$ with b even, the valid pairs are

$$(b, a) = (6, 9), (8, 7),$$

giving 2 pairs. Step 8: Therefore the total number of valid numbers is

$$4 + 2 = 6.$$

Q17

Step 1: The capsule consists of a cylinder and two hemispheres, which together make one sphere. Its surface area is

$$S = 2\pi rh + 4\pi r^2.$$

Step 2: When all linear dimensions are doubled, surface area becomes four times the original. Hence the increase is

$$4S - S = 3S.$$

Given $3S = 312\pi$, we get

$$S = 104\pi.$$

Thus

$$2\pi rh + 4\pi r^2 = 104\pi,$$

or

$$rh + 2r^2 = 52.$$

Step 3: The volume of the capsule is

$$V = \pi r^2 h + \frac{4}{3}\pi r^3.$$

Step 4: When all linear dimensions are doubled, volume becomes eight times the original. Hence the increase is

$$8V - V = 7V.$$

Given

$$7V = \frac{3472\pi}{3},$$

we get

$$V = \frac{496\pi}{3}.$$

Step 5: Therefore,

$$\pi r^2 h + \frac{4}{3}\pi r^3 = \frac{496\pi}{3}.$$

Cancel π :

$$r^2 h + \frac{4}{3}r^3 = \frac{496}{3}.$$

Step 6: From

$$rh + 2r^2 = 52,$$

multiply by r :

$$r^2h + 2r^3 = 52r.$$

Substitute $r^2h = 52r - 2r^3$ into the volume equation:

$$52r - 2r^3 + \frac{4}{3}r^3 = \frac{496}{3}.$$

Step 7: Multiply by 3:

$$156r - 6r^3 + 4r^3 = 496,$$

so

$$r^3 - 78r + 248 = 0.$$

Testing the positive integral factor gives

$$r = 4.$$

Step 8: Use

$$rh + 2r^2 = 52.$$

With $r = 4$,

$$4h + 32 = 52,$$

so

$$h = 5.$$

Therefore,

$$h - r = 5 - 4 = 1.$$

Q18

Step 1: Since diagonals of a parallelogram bisect each other, the midpoint of AC is

$$\left(\frac{p}{2}, \frac{q+1}{2}\right).$$

Step 2: This midpoint lies on $2x + y = 21$, so

$$2\left(\frac{p}{2}\right) + \frac{q+1}{2} = 21.$$

Thus

$$p + \frac{q+1}{2} = 21.$$

Step 3: Multiplying by 2,

$$2p + q + 1 = 42,$$

so

$$2p + q = 41.$$

Step 4: The adjacent side vectors are

$$\overrightarrow{AB} = (6, 3)$$

and

$$\overrightarrow{BC} = (p - 6, q - 4).$$

The area of the parallelogram is

$$|6(q - 4) - 3(p - 6)| = 45.$$

Step 5: Simplify:

$$|6q - 24 - 3p + 18| = 45,$$

so

$$|6q - 3p - 6| = 45.$$

Dividing by 3,

$$|2q - p - 2| = 15.$$

Step 6: From $2p + q = 41$,

$$p = \frac{41 - q}{2}.$$

Substitute into the area condition:

$$\left| 2q - \frac{41 - q}{2} - 2 \right| = 15.$$

This gives

$$\left| \frac{5q - 45}{2} \right| = 15.$$

Step 7: Hence

$$|5q - 45| = 30,$$

so

$$q = 15 \quad \text{or} \quad q = 3.$$

Step 8: The condition $q < 5$ selects $q = 3$. Then

$$2p + 3 = 41,$$

so

$$p = 19.$$

Therefore,

$$p + q = 19 + 3 = 22.$$

Q19

Step 1: Since D is midpoint of AB and AE:EC=2:1, triangle ADE has sides along the two arms of triangle ABC. Step 2: $\text{Area}(\text{ADE})/\text{area}(\text{ABC}) = (\text{AD}/\text{AB})(\text{AE}/\text{AC}) = (1/2)(2/3) = 1/3$. Step 3: Use coordinates A=(0,0), B=(2,0), C=(0,3), so D=(1,0) and E=(0,2). Step 4: Line DE meets the line through C parallel to AB at F=(-1/2,3). Step 5: Thus triangle CEF has base 1/2 and height 1, while triangle ADE has area 1 in the scaled figure, so $\text{area}(\text{CEF})=18/4=4.5$.

Q20

Step 1: The final enlarged area is $3 \times 45 = 135$. Step 2: The length-breadth changes multiply area by $1.25 \times 0.90 = 1.125$. Step 3: Let original area be A. Step 4: Then $1.125A=135$. Step 5: Therefore $A=120$ square cm.

Q21

Step 1: Counts of multiples of 6,10,15 are 100,60,40. Step 2: Pairwise lcm counts are $\text{lcm}(6,10)=30$ gives 20, $\text{lcm}(6,15)=30$ gives 20, and $\text{lcm}(10,15)=30$ gives 20. Step 3: Triple lcm is 30, count 20. Step 4: Exactly one count is singles -2(pairwise overlaps) +3(triple overlaps). Step 5: So $200 - 120 + 60 = 140$.

Q22

Step 1: Multiples are 75,50,30. Step 2: Pairwise lcm counts are 25,15,10. Step 3: Triple lcm 60 gives 5. Step 4: Exactly one = singles -2(pair overlaps) +3(triple). Step 5: $155-100+15=70$.

Evidence Lineage

Q	Candidate	Source Question IDs
1	CAT_QA_SPEC_07_CCAUD2017S1_QA_Q16;CAT_2018_S1_QA_Q09	
2	CAT_QA_SPEC_18_CCAUD2017S1_QA_Q03;CAT_2019_S1_QA_Q03	
3	CAT_QA_SPEC_19_CCAUD2017S1_QA_Q19;CAT_2017_S1_QA_Q19	
4	CAT_QA_SPEC_20_CCAUD2017S1_QA_Q13;CAT_2022_S2_QA_Q13	
5	CAT_QA_SPEC_17_CCAUD2018S2_QA_Q26;CAT_2020_S1_QA_Q01	
6	CAT_QA_SPEC_01_ECANE2017S1_QA_Q12;CAT_2017_S2_QA_Q07	
7	CAT_QA_SPEC_10_CCAUD2018S1_QA_Q01;CAT_2018_S1_QA_Q14	
8	CAT_QA_SPEC_06_ECANE2017S1_QA_Q01;CAT_2017_S1_QA_Q05	
9	CAT_QA_SPEC_03_ECANE2017S1_QA_Q13;CAT_2018_S1_QA_Q26	
10	CAT_QA_SPEC_02_CCAUD2017S1_QA_Q34;CAT_2017_S2_QA_Q34	
11	CAT_QA_SPEC_15_CCAUD2017S1_QA_Q26;CAT_2017_S2_QA_Q26	
12	CAT_QA_SPEC_08_CCAUD2017S1_QA_Q03;CAT_2017_S1_QA_Q26	
13	CAT_QA_SPEC_12_CCAUD2018S2_QA_Q28;CAT_2018_S2_QA_Q29	
14	CAT_QA_SPEC_13_CCAUD2018S2_QA_Q30;CAT_2019_S1_QA_Q09	

Q	Candidate	Source Question IDs
15	CAT_QA_SPEC_07_CCAUDE10	CAT_2018_S1_QA_Q16;CAT_2017_S1_QA_Q29
16	CAT_QA_SPEC_20_GPAT552019	CAT_2019_S2_QAG106;CAT_2022_S2_QA_Q13
17	CAT_QA_SPEC_19_GPAT552017	CAT_2017_S1_QA_Q19
18	CAT_QA_SPEC_18_GPAT552017	COORDINATES 2019_S1_QA_Q03
19	CAT_QA_SPEC_07_ECAHE2004	CAT_2017_S1_QA_Q16;CAT_2017_S1_QA_Q20
20	CAT_QA_SPEC_19_ECAHE2004	CAT_2017_S1_QA_Q18;CAT_2017_S1_QA_Q19
21	CAT_QA_SPEC_20_ECAHE2004	CAT_2019_S2_QA_Q06;CAT_2022_S2_QA_Q13
22	CAT_QA_SPEC_20_FCAT_00	CAT_2019_S2_QA_Q06;CAT_2022_S2_QA_Q13